

Application of **Image Processing** **Technology in VLPR**



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Abstract

- VLPR = number detection & recognition of vehicle plates
- Combines: Digital Image Processing + Machine Learning
- Key stages: preprocessing → detection → segmentation → classification
- Applications: toll collection, parking, surveillance



SURVEILLANCE



VEHICLE

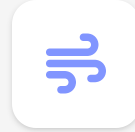


PLATE

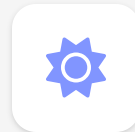
Introduction

- Part of Intelligent Transportation Systems (ITS)
- Used for automation and monitoring
- **Need:** robust preprocessing + feature extraction

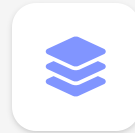
CHALLENGES



Noise, blur, shadows



Lighting variations



Complex backgrounds

Problem Statement & Objectives

- Problem: manual monitoring is inefficient
 - Need: automated license plate recognition
-

SYSTEM GOALS

OBJECTIVES

01

Detect plate accurately

02

Segment characters correctly

03

Recognize characters using ML

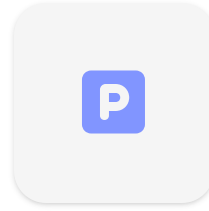
04

Ensure fast and efficient processing

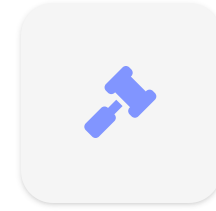
Applications of VLPR



Toll collection



**Parking management
systems**



**Traffic monitoring & law
enforcement**



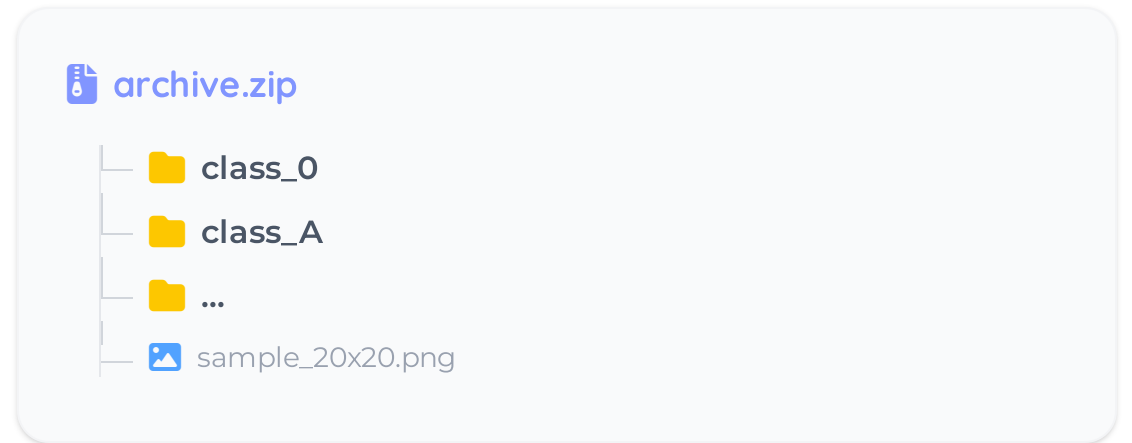
**Security & surveillance
systems**



Smart city infrastructure

Dataset Description

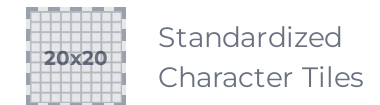
- Labeled dataset: digits 0–9, alphabets A–Z
- Stored in archive.zip (folder-wise classification)
- Each class → separate directory
- Images resized to 20×20 pixels
- Helps model learn variations in shape and style



DATASET SPLIT



RESOLUTION



System Workflow

Comprehensive VLPR Pipeline Flowchart

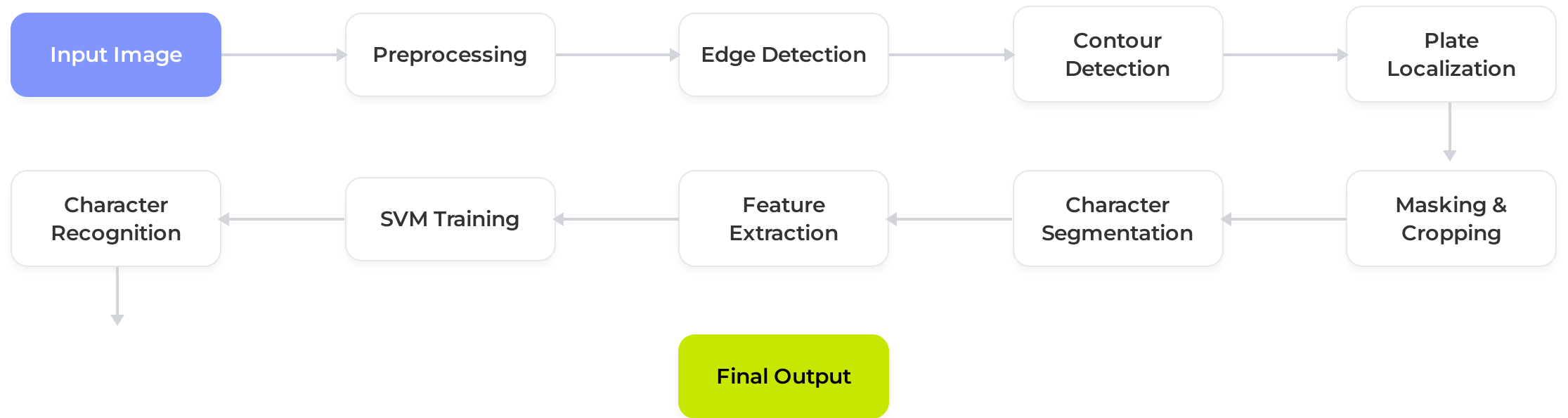


Image Preprocessing



Step 01

Convert RGB → Grayscale



Step 02

Apply Gaussian Filter
(noise reduction)



Step 03

Otsu Thresholding
for binarization

✓ Benefits

- Improves contrast
- Reduces noise
- Simplifies image processing

Edge Detection & Plate Localization

- Canny Edge Detection highlights object boundaries
- Contour detection used to find shapes
- Sort contours by area
- Polygon approximation → detect rectangles

Plate identified as a 4-sided contour

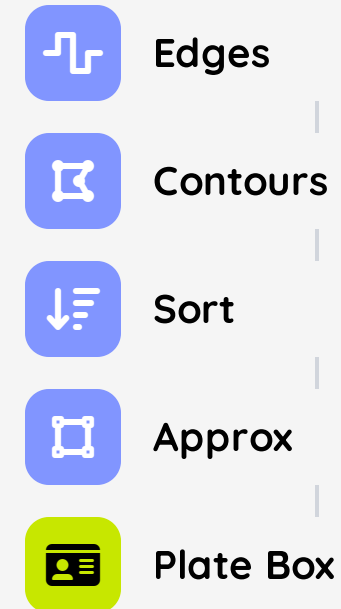
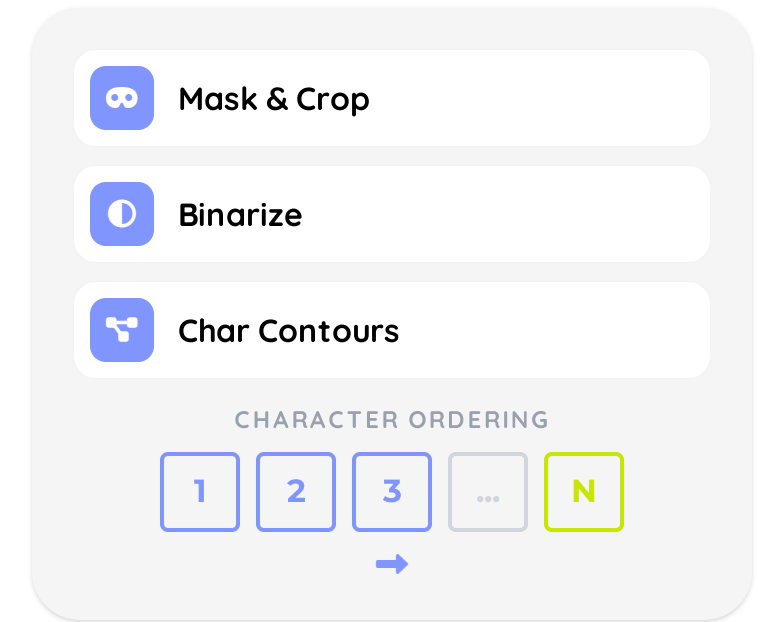


Plate Extraction & Segmentation

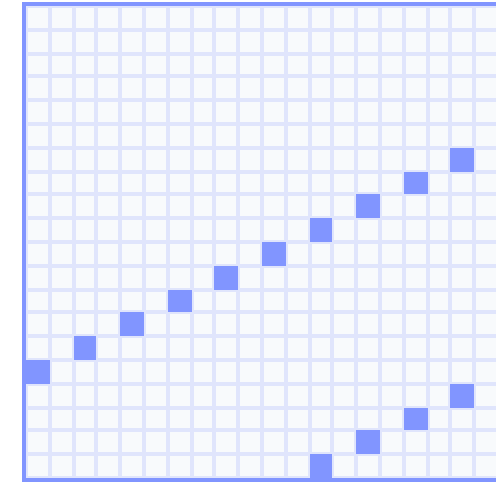
- Create mask to isolate plate region
- Crop the plate from image
- Apply binarization for clarity
- Contour detection for characters
 - ▼ Filter characters by: Size, Aspect ratio
 - ↕ Sort characters left → right



Feature Extraction

- Resize each character → 20×20 pixels
- Convert to 1D vector (400 features)
- Pixel intensity values used as features
- Normalize values to 0–1

Advantages: simple, fast, efficient



20 × 20 Grid



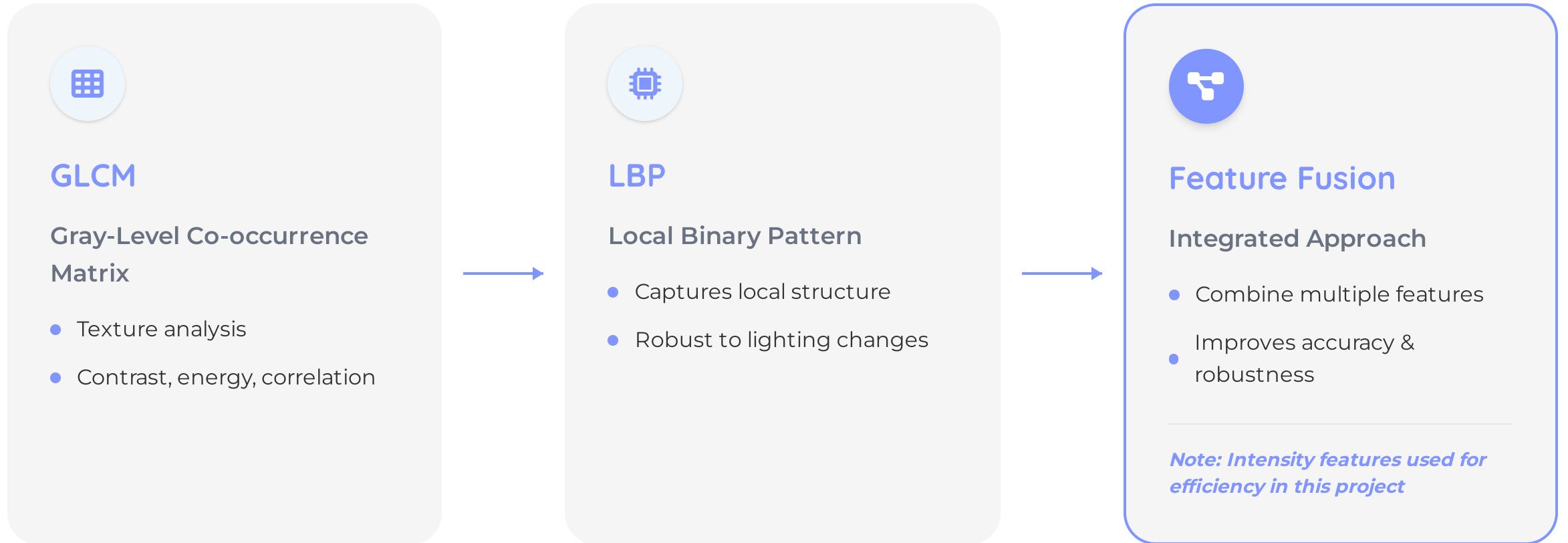
FLATTEN & NORMALIZE



400-Length Vector

(Values: 0.0 — 1.0)

Advanced Feature Techniques



Model Training (SVM)



- Use Support Vector Machine (SVM)
- Linear kernel used for efficiency
- Input: feature vectors
- Output: predicted characters

★ Advantages

- ✓ High accuracy for classification
- ✓ Works well with high-dimensional data

Experimental Setup & Results

Setup & Conditions

- Tested under:
 -  Different lighting conditions
 -  Various angles & backgrounds

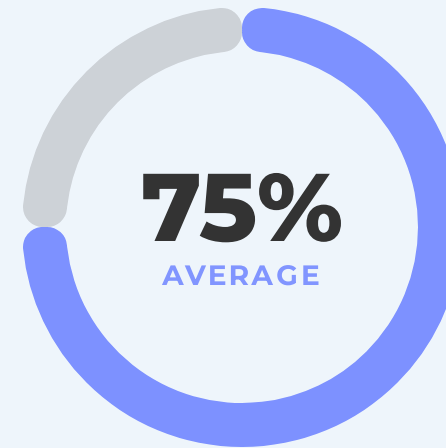
Strengths:

- Fast processing
- Good detection accuracy

Limitations:

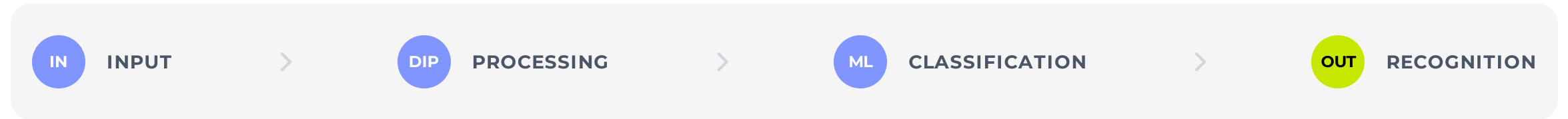
- Blur/noise reduce performance
- Similar characters confusion (O/O, B/8)

Performance Accuracy



Accuracy Range: ~70–80%

Conclusion & Future Scope



✓ Conclusion

- VLPR system successfully implemented
- Combines DIP + ML effectively
- Efficient and practical for real-time use

🚀 Future Improvements



Deep Learning (CNN, YOLO)



Improve segmentation accuracy



Use feature fusion techniques



Handle low-light and complex environments